

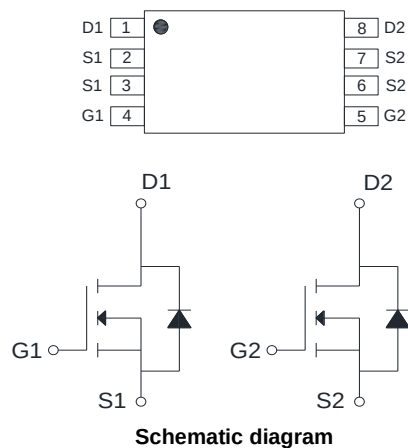
Features

- $V_{DS}=20V$, $I_D=5A$
- $R_{DS(ON)}=18m\Omega@V_{GS}=4.5V(Typ.)$
- $R_{DS(ON)}=24m\Omega@V_{GS}=2.5V(Typ.)$
- High Power and current handling capability
- Lead free product is acquired
- Surface Mount Package

Main Applications

- Battery Protection
- Load Switch
- Power Management

TSSOP8



Absolute Maximum Ratings ($T_{Ambient}=25^{\circ}C$ unless noted otherwise)

Symbol	Description	Limit	Unit
V_{DS}	Drain-Source Voltage	20	Volt
V_{GS}	Gate-Source Voltage	± 12	Volt
I_D	Drain Current-Continuous	5	Amp
I_{DM}	Drain Current-Pulsed ^(Note 1)	25	Amp
P_D	Maximum Power Dissipation	1.4	Watt
T_J	Operating Temperature Range	-55 to +150	$^{\circ}C$
T_{STG}	Storage Temperature Range	-55 to +150	$^{\circ}C$

Thermal Characteristic

Symbol	Description	Limit	Unit
$R_{\theta JA}$	Thermal Resistance, Junction to Ambient ^(Note 2)	125	$^{\circ}C/W$

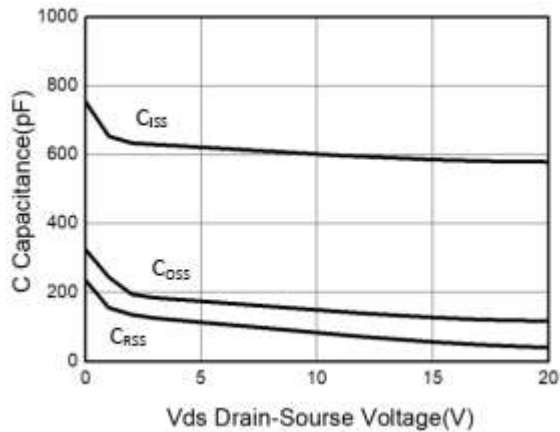
Electrical Characteristics ($T_{Ambient}=25^{\circ}C$ unless noted otherwise)

Symbol	Parameters	Conditions	Min.	Typ.	Max.	Unit
Off Characteristics						
BV _{DSS}	Drain-Source Breakdown Voltage	V _{GS} =0V, I _D =250μA	20			Volt
I _{DSS}	Zero Gate Voltage Drain Current	V _{DS} =20V, V _{GS} =0V			1	μA
I _{GSS}	Gate-Body Leakage Current	V _{GS} =±12V, V _{DS} =0V			±100	nA
On Characteristics(Note 3)						
V _{GS(th)}	Gate Threshold Voltage	V _{DS} = V _{GS} , I _D =250μA	0.5	0.7	1.2	Volt
R _{DS(ON)}	Drain-Source On-State Resistance	V _{GS} =4.5V, I _D =5A		18	24	mΩ
		V _{GS} =2.5V, I _D =4A		24	30	mΩ
Dynamic Characteristics(Note 4)						
C _{iss}	Input Capacitance	V _{DD} =10V, V _{GS} =0V , F=1MHz		550		pF
C _{oss}	Output Capacitance			125		pF
C _{rss}	Reverse Transfer Capacitance			64		pF
Switching Characteristics(Note 4)						
td(on)	Turn-on Delay Time	V _{DD} =10V, I _D =5A V _{GS} =4.5V, R _{GEN} =10Ω		9		nS
tr	Turn-on Rise Time			10		nS
td(off)	Turn-Off Delay Time			32		nS
tf	Turn-Off Fall Time			24		nS
Q _g	Total Gate Charge	V _{DS} =10V, I _D =5A, V _{GS} =4.5V		9.5		nC
Q _{gs}	Gate-Source Charge			2.1		nC
Q _{gd}	Gate-Drain Charge			1.4		nC
Drain-Source Diode Characteristics						
V _{SD}	Diode Forward Voltage ^(Note 3)	V _{GS} =0V, I _S =2.0A			1.16	V

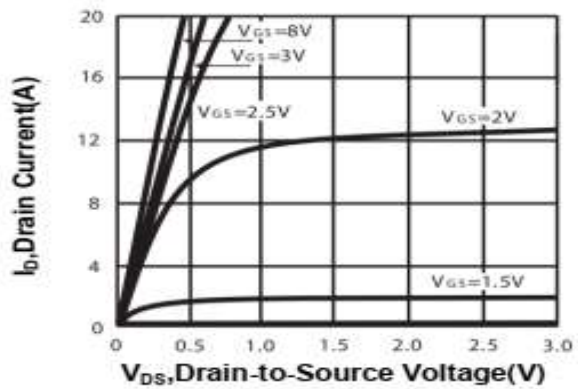
- Notes:**
1. Repetitive Rating: Pulse width limited by maximum junction temperature
 2. Surface mounted on FR4 Board, $t \leq 10$ sec
 3. Pulse Test: Pulse Width $\leq 300\mu s$, Duty Cycle $\leq 2\%$
 4. Guaranteed by design, not subject to production

Typical Electrical and Thermal Characteristics

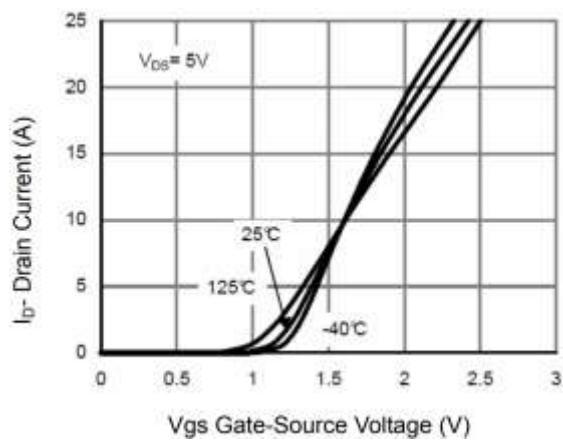
Capacitance Characteristic



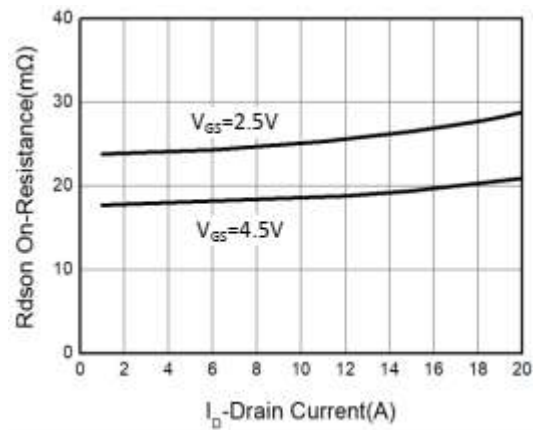
Output Characteristic



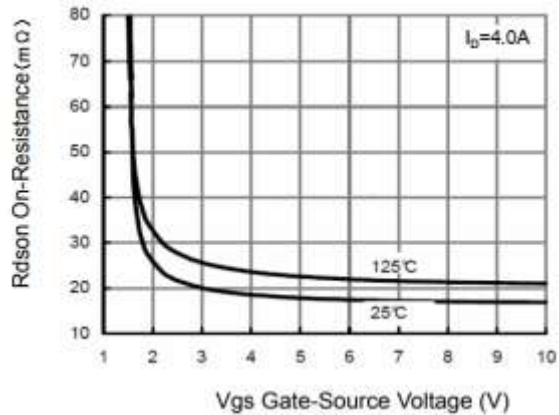
Transfer Characteristic



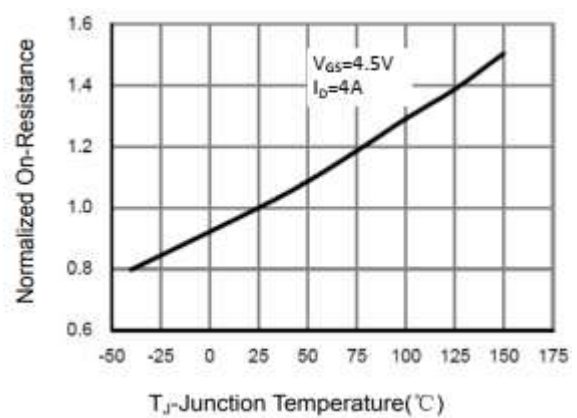
On-Resistance vs. Drain Current



On-Resistance vs. Vgs



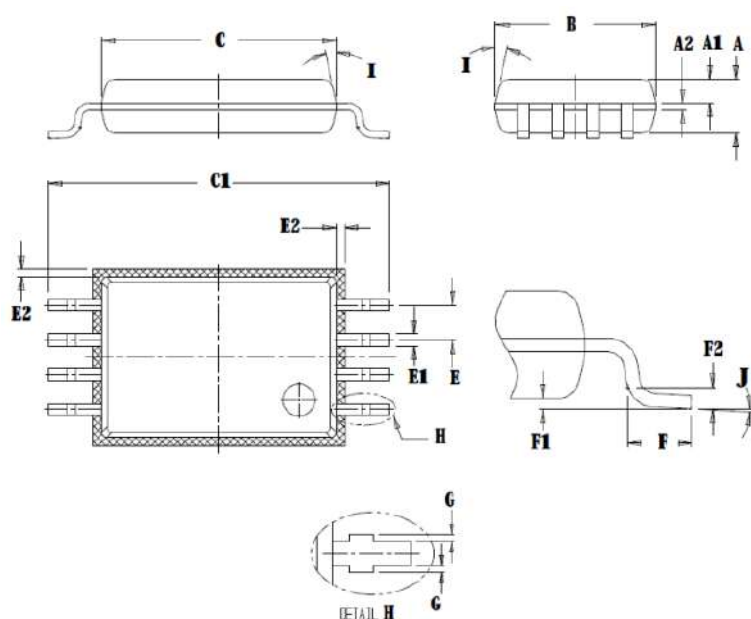
On-Resistance vs. Junction Temperature



PACKAGE OUTLINE

Plastic surface mounted package; 8 leads

TSSOP8



COMMON DIMENSIONS UNITS MEASURE=MILLIMETER			
SYMBOL	MIN	MID	MAX
A	0.95	1.00	1.05
A1	0.39	0.44	0.49
A2	—	0.127	—
B	2.95	3.00	3.05
C	4.35	4.40	4.45
C1	6.30	6.40	6.50
E	—	0.65 TYP	—
E1	0.195	0.22	0.245
E2	—	0.12	—
F	0.5	0.60	0.7
F1	0	0.05	0.1
F2	—	0.2	—
G	—	0.075	—
I	10 °	12 °	14 °
J	0 °	3 °	6 °